

Best Locations for a New Healthy Food Store in Philadelphia

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Introduction:

The overall wellbeing of a community dependent on their access to healthy and affordable foods, yet many areas in Philadelphia lack this essential. And this issue goes beyond Philadelphia, according to the Food Empowerment Project¹, major cities like Chicago, Los Angeles, and New York City are also struggling with similar problems of areas having limited access to fresh vegetables, fruits, and other nutritious foods. To solve this issue, they are implementing local initiatives to resolve it because if the issue is kept, major health problems may occur, those includes obesity, diabetes, and other diet related problems². For this, the city government at Philadelphia is also looking to encourage the department of new stores that sells healthy food in a target location. The goal is to place the stores in an area that maximizes accessibility while also supporting community investments. The store should be located in a Philadelphia empowerment zone, within 2,000 feet of a subway or regional rail station, at least 1,200 feet way from existing farmer market or corner stores, and situated in a large contiguous area.

Data and Methods:

I begin by downloading the publicly available data from PASDA³, the Pennsylvania Geospatial Clearinghouse of Philadelphia empowerment zones, of subway stations, regional rail station, farmers market, and corner stores involved in the Healthy Concern Store Initiative. I then opened and imported the data into ArcGIS Pro. I then made sure that all the layers and my map were in State Plane Pennsylvania South CRS. First, I created a 2,000ft buffer of the regional rail and the subway stations by going into the Analysis tab and then selecting Piecewise Buffer from the Analysis Gallery option. For input feature, I selected the regional rail and then I entered 2,000 for distance with unit as US Survey feet with dissolved all. I repeated this same process for the subway stations. I then did a union of the resulting regional rail station buffer with the subway station buffer by selecting union from the Analysis Gallery option. The input features were the regional rail and subway stations buffer. Then using the Piecewise Intersect from the Analysis Gallery, I intersected the regional rail and subway station buffer union layer with the empowerment zones and named it inclusion. They were also the input features. I then created an exclusion layer of where the new stores cannot be located. For this, buffered the farmers markets and the corners stores following the same methods as before, with 1,200ft buffer. But this time, the intersection of these layer with empower zones having been named exclusive.

¹ Food Empowerment Project. "Food Deserts" Accessed March 15, 2025. <https://foodispower.org/access-health/food-deserts/>

² Medical News Today. "What are food deserts, and how do they impact health?" Accessed March 15, 2025. <https://www.medicalnewstoday.com/articles/what-are-food-deserts>

³ PASDA. "Pennsylvania Spatial Data Access" Accessed March 15, 2025. <https://www.pasda.psu.edu/>

I then created a layer of the candidate areas where the new stores must be located but without the areas where they cannot be. I did this by removing the exclusive layer from the inclusive layer using the Erase function from the Analysis Gallery. I turned this new layer into each of the polygon having represented by one record in the Attribute Table using the Multipart to Singlepart operations from the search bar. I then opened the Attribute Table for this new layer and added a field for area with data type as double. I did a right click on the Area field in the Attribute Table then selected the Calculate Geometry. I selected the input feature as the most recent layer that was converted from Multipart to Singlepart. Then the field being Area and property being Area with area unit being US Survey Acres. The coordinate system being State Plane Pennsylvania South CRS. I then clicked the apply button to calculate the area of each part of the separate polygons. In the end, I made maps of the three major processes, with the final having graduated color map of the field as area with the method as manual. I then changed the color to red and green and lowered the green bound to anything 1 or higher.

Results:

My findings from the creation of the inclusive layer of where the new store must be located, figure 1, show that there were many areas within the empower zone where the new store was within 2,000 feet from a subway or regional rail station. But when I made the exclusive layer, figure 2, there were also many areas within the 1,200 feet buffer, where there was a farmer market or corner store participating in the Healthy Corner Store Initiative. I removed the exclusive layer from the inclusive layer, figure 3, to show where there were no farmer market or concern store participating in the Healthy Corner Store Initiative and public transit was available. There were 5 patches of areas where the store could be placed. The areas of those patches were 92.98, 25.01, 1.03, 0.40, and 0.09 Acres as shown in figure 4. Of those only 3 of them were at least 1 acre or more and 2 of them were not.

Discussion:

I used figure 1 and 2 to show the process I took to get to figure 3 outputs. In each of those, I made sure to show the outline of the Empowerment Zone to make it visually comparable. In figure 3, I also highlighted the large areas where a new store selling healthy food could be built using the graduated color map. There were five patches, 92.98, 25.01, 1.03, 0.40, and 0.09 acres. This new store could be built in the patch of location where the area is higher than 1 acre. It means there are three possible options. There were two areas that were less than 1 acre and were hard to see on the map. In figure 4, I included the Attribute table for figure 3 to indicate the area of the patches. When considering which area matches which patch, it can be figured out based on the size of their shape; they are somewhat differentiable. With 92.98 acres being the largest and 0.09 acres being the smallest patch. 1.03 acres being the smallest patch where the new store could be built.

Conclusion:

The Empowerment Zone has five areas where the specific spatial criteria are met, but there are only three areas where the store could be approved to be built. The highest area patch has 92.98 acres and lowest patch, where it could be built, having 1.03 acres. Since the highest area has an increased likelihood of securing the potential property, it would most likely be approved to be built at either the 92.98 acres area or the 25.01 acres area. These areas are both within the 2,000 feet of a subway or regional rail station and not within 1,200 feet of a farmers' market or corner store participating in the Healthy Corner Initiatives. Overall, the highest areas location, 92.98 acres, have higher likelihood of securing a potential property.

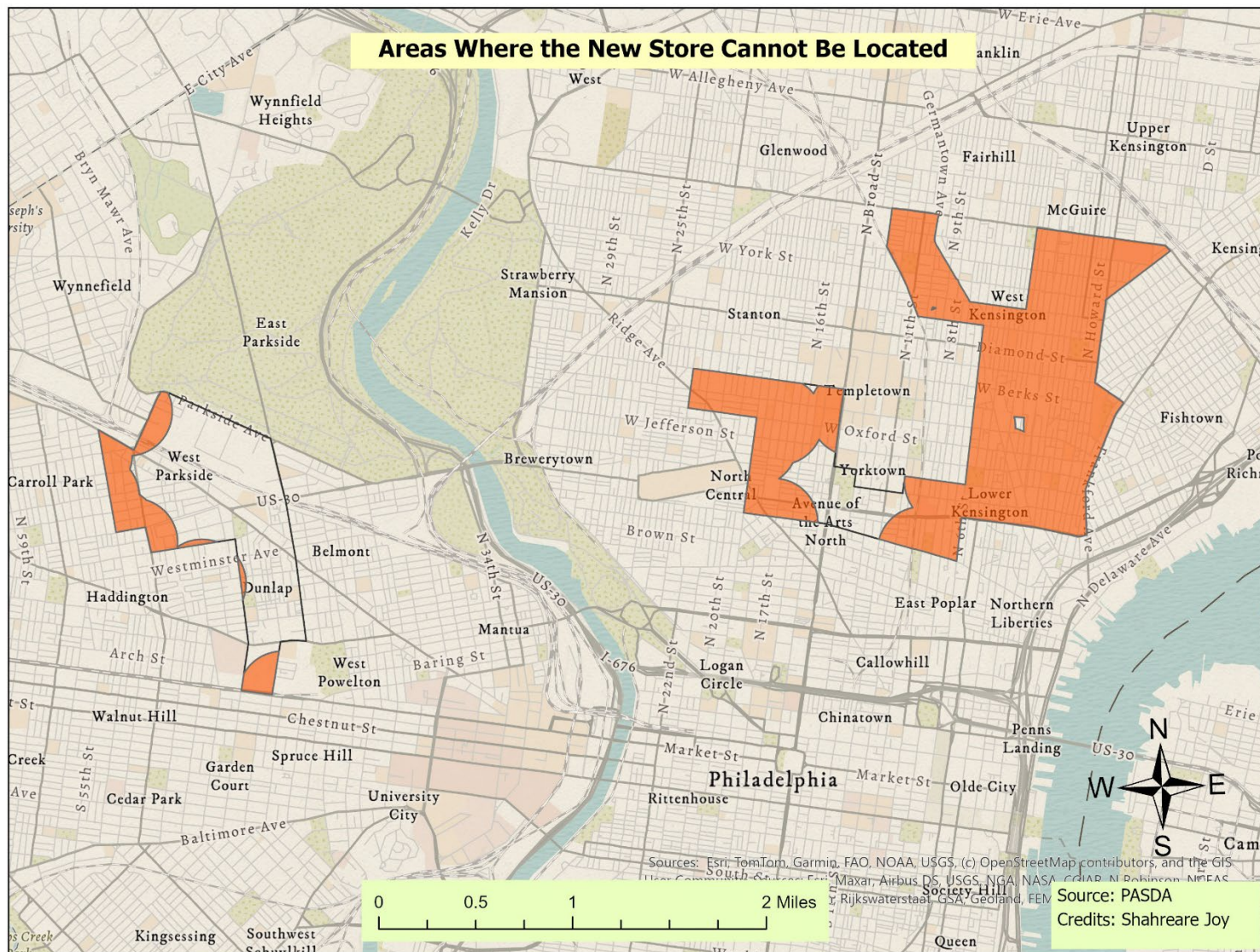


Figure 2: A map showing the areas, in red, that are within the Philadelphia Empowerment Zone that are within 1200 feet of a farmers' market or corner store participating in the Healthy Corner Store Initiative.

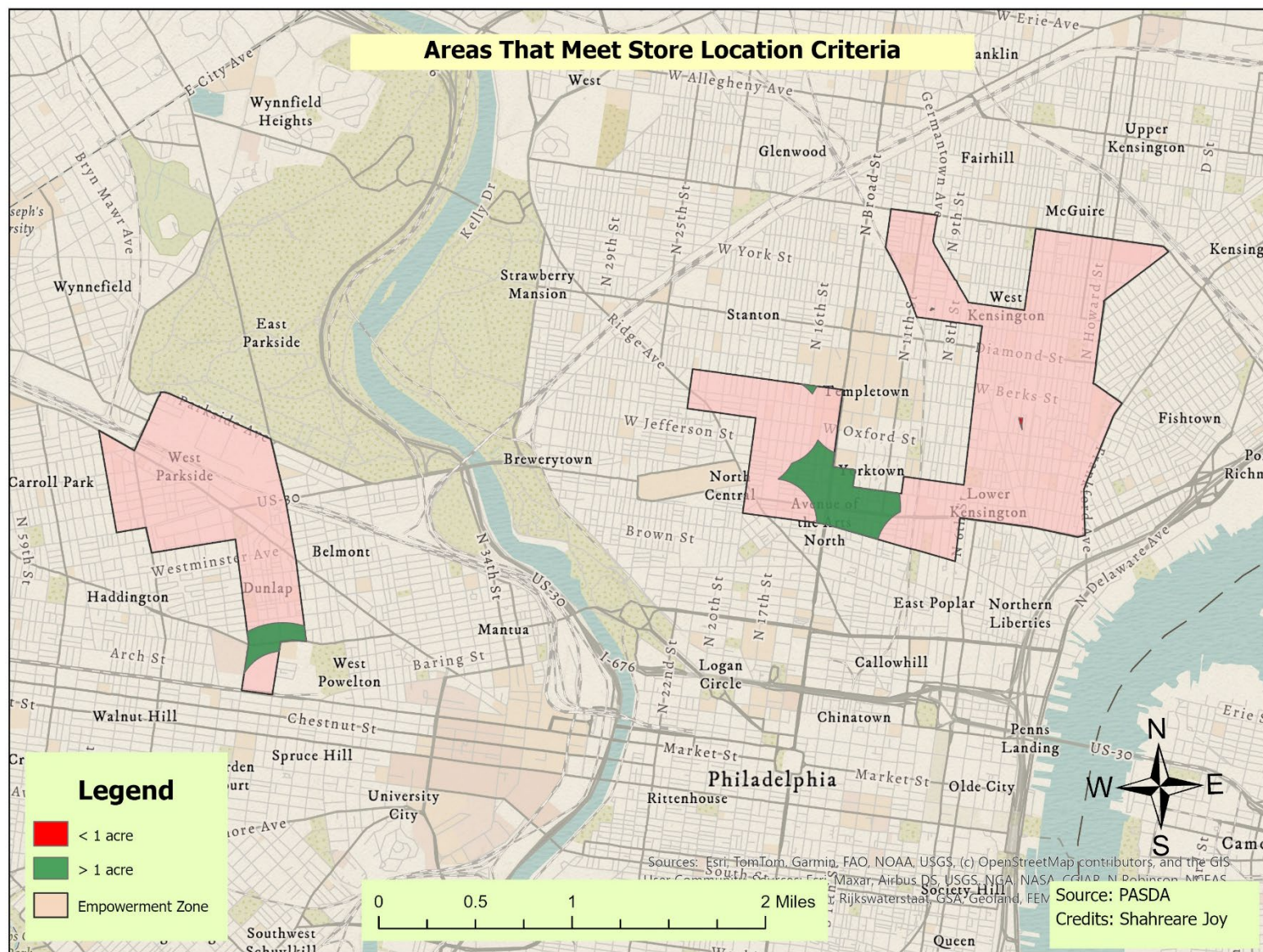


Figure 3: A map showing the areas within the Philadelphia Empowerment Zone, where a new store selling healthy foods can be built.

	OBJECTID *	Shape *	FID_r...	FID_r...	FID_r...	FID_P...	PERIMETER	EMPZN_	EMPZN_ID	Areaft	Shape_Length	Shape_Area	Area_Acre
1	1	Polygon	1	1	-1	0	57053.464837	2	2	47201790.6489	9867.633389	4050279.251184	92.981617
2	2	Polygon	1	1	-1	0	57053.464837	2	2	47201790.6489	794.452228	17348.976005	0.398278
3	3	Polygon	1	1	-1	0	57053.464837	2	2	47201790.6489	1103.519253	44654.926199	1.025136
4	4	Polygon	1	1	-1	1	25602.515909	3	1	21087294.6462	5334.90861	1089437.172324	25.010036
5	5	Polygon	2	-1	1	0	57053.464837	2	2	47201790.6489	312.848871	3841.572456	0.08819

Figure 4: A screenshot showing the attribute table of figure 3, with the area values of each polygon.